

POWERCO CUSTOMER CHURN

Executive Summary • September 2026

Churn affects ~1 in 10 customers, but the current model needs improvement before it can reliably support retention decisions.

KEY FINDINGS

- **Scale:** 14,606 customers analysed; 1,419 churned (9.72% churn rate).
- **Drivers:** Consumption, product usage, margins, and pricing changes identified as key dimensions.
- **Price Sensitivity:** An important hypothesis, though not established as the sole driver.

MODEL PERFORMANCE (RANDOM FOREST)

90.31%

ACCURACY

71.43%

PRECISION

5.46%

RECALL

0.665

ROC-AUC

CRITICAL LIMITATION: The model identifies only 20 out of 366 actual churners (5.46% Recall). It should NOT be used as a standalone tool for automated retention decisions.

RECOMMENDATIONS & NEXT STEPS

- **Addressing Class Imbalance:** Improve churn detection and optimise prediction thresholds.
- **Segmentation:** Investigate pricing/consumption to find high-risk segments.
- **Targeting:** Shift from broad discounts to risk-based retention strategies.
- **Validation:** Test improved models on future data before operational deployment.

BUSINESS IMPACT

A stronger prediction model will allow PowerCo to identify at-risk customers early, allowing for prioritized resource allocation and reduction of avoidable losses.